

Optimization Techniques in Power System (BT EE610)



हरियाणा केंद्रीय विश्वविद्यालय
CENTRAL UNIVERSITY OF HARYANA
NAAC ACCREDITED 'A' GRADE UNIVERSITY

Unit-I Design Optimization

- ❖ What is Design Optimization?
- ❖ Types of Optimization Algorithm
- ❖ Optimal Problem formulation
- ❖ Design Variables
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- ❖ Objective functions
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By
Dr. Manish Kumar
Asth. Professor
Department of Electrical Engineering
SOE&T, CUH

What is Design optimization?

1.0. Introduction:

In optimization of a design, the design objective could be simply to minimize the cost of production or to maximize the efficiency of production. An optimization algorithm is a procedure which is executed iteratively by comparing various solutions till an optimum or a satisfactory solution is found.

With the advent of computers, optimization has become a part of computer-aided design activities. There are two distinct types of optimization algorithms widely used today.

(a) Deterministic Algorithms.

They use specific rules for moving one solution to other. These algorithms are in use to suite some times and have been successfully applied for many engineering design problems.

(b) Stochastic Algorithms.

The stochastic algorithms are in nature with probabilistic translation rules. These are gaining popularity due to certain properties which deterministic algorithms do not have.

2.0 Optimal problem formulation:

A naive optimal design is achieved by comparing a few (limited up to ten or so) alternative solutions created by using a priori problem knowledge. In this method feasibility of each design solution is first investigated. Thereafter an estimate of underlying objective (cost, profit, etc.,) of each solution is compared and best solution is adopted.

It is impossible to apply single formulation procedure for all engineering design problems, since the objective in a design problem and associated therefore, design parameters vary product to product different techniques are used in

different problems. Purpose of formulation is to create a mathematical model of the optimal design problem, which then can be solved using an optimization algorithm. Figure 1 shows an outline of the steps usually involved in an optimal design formulation.

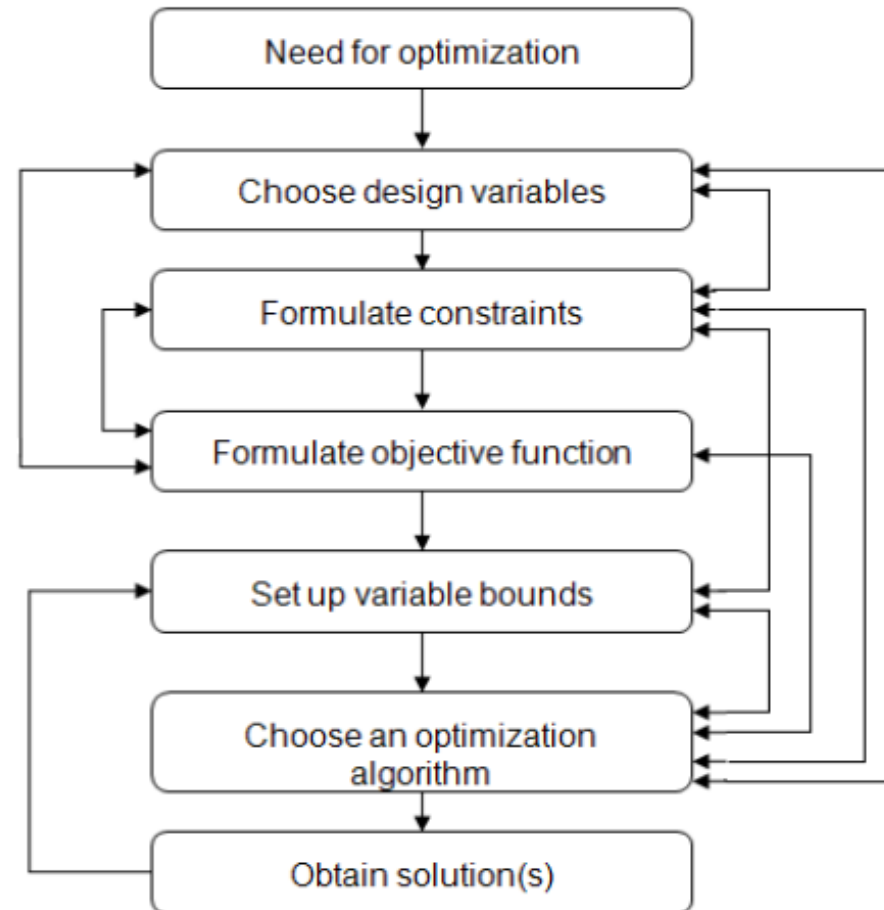


Fig.1 A flowchart of the optimal design procedure.

Design variables:

The formulation of an optimization problem begins with identifying the underlying design variables, which are primarily varied during the optimization process. A design problem usually involves many design parameters, of which some are highly sensitive to the proper working of the design. These parameters are called design variables in the parlance of optimization procedures. Other (not so important) design parameters usually remain fixed or vary in relation to the design variables.

The first thumb rule of the formulation of an optimization problem is to choose as few design variables as possible. The outcome of that optimization procedure may indicate whether to include more design variables in a revised formulation or to replace some previously considered design variables with new design variables.

Constraints:

The constraints represent some functional relationships among the design variables and other design parameters satisfying certain physical phenomenon and certain resource limitations. The nature and number of constraints to be included in the formulation depend on the user. Constraints may have exact mathematical expressions or not.

For example, maximum stress is a constraint of a structure. If a structure has regular shape they have an exact mathematical relation of maximum stress with dimensions. But incase irregular shape, finite element simulation software may be necessary to compute the maximum stress.

The following two types of constraints emerge from most considerations:

1. Inequality type constraints.
2. Equality type constraints.

Inequality constraints state that the functional relationships among variables are either greater than, smaller than or equal to, a resource value.

Example:

The stress $\sigma(x)$ developed anywhere in a component must be smaller than or equal to the allowable strength ($S_{allowable}$) of the material.

$$\sigma(x) \leq S_{allowable}$$

Some constraints may be of greater-than / equal-to type. For example, the natural frequency ($f(x)$) of a system to be greater than 2 Hz or by notation $f(x) \geq 2$.

Equality constraints state that functional relationships should exactly match a resource value.

Example:

The deflection $\delta(x)$ of a point in the component must be exactly equal to 5 mm. Then $\delta(x) = 5$.

It is very difficult to handle the equality constraints in the algorithms. In such cases, equality constraint is relaxed by including two inequality constraints as given below.

Example:

Previously $\delta(x) = 5$

Now it is changed to inequality constraints as given below:

$$\delta(x) \geq 4,$$

$$\delta(x) \leq 6.$$

Objective functions:

The next task in the formulation procedure is to find the objective function in terms of the design variables and other problem parameters. The common engineering objectives involve minimization of overall cost of manufacturing or minimization of overall weight of a component or maximization of total life of a product or others.

Although most of the objectives can be quantified (expressed in mathematical form), there are some objectives (such as aesthetic aspect of a design, ride characteristics of a car suspension design and reliability of a design) that may not be possible to formulate mathematically. In such a case an approximating mathematical expression is used.

In real world optimization, there could be more than one objective that the designer may want to optimize simultaneously. The multiple objective optimization algorithms are complex and computationally expensive. Therefore the most important objective is chosen as the objective function and the other objectives are included as constraints by restricting their values within a certain range.

For example, consider optimal truss structure design problem. The designer may be interested in minimizing the overall weight of the structure and simultaneously be concerned in minimizing the deflection of a specific point in the truss. In the optimal problem formulation, the designer may like to use the weight of the truss (as a function of the cross sections of the members) as the objective function and have a constraint with the deflection of the concerned point to be less than a specific limit.

The objective function can be of two types. Either it is to be maximized or it has to be minimized. Usually the optimization algorithms were written for minimization problems or maximization problems. Although in some algorithms, some minor structural changes would enable to perform either minimization (or) maximization; this requires extensive knowledge of the algorithm.

The duality principle helps by allowing the same algorithm to be used for minimization or maximization with a minor change in the objective function instead of a change in the entire algorithm. If the algorithm is for solving a minimization problem, it can be easily changed to a maximization problem by multiplying the objective function by -1 and vice versa.

Variable bounds:

The final task of the formulation procedure is to set the minimum and the maximum bounds on each design variable. Certain optimization algorithms do not require this information. In these problems, the constraints completely surround the feasible region. Other problems require the search algorithm with in these bounds.

In general, all N design variables are restricted to lie within the minimum and the maximum bounds as follows.

$$x_i^{(L)} \leq x_i \leq x_i^{(U)} \quad \text{for } i = 1, 2, 3, \dots N. \quad (1)$$

In any given problem, the determination of the variables bounds $x_i^{(L)}$ and $x_i^{(U)}$ may be difficult. One way to remedy this situation is to make a guess about the optimal solution and set the minimum and maximum bounds so that the optimal solution lies within these two bounds

If any design variable corresponding to the optimal solution is found to lie on or near the minimum or maximum bound, the chosen bound may be adjusted and optimization algorithm may be simulated again.

After the above four tasks are completed, the optimization problem can be mathematically written in a special format, known as nonlinear programming (NLP) format.

General format:

Denoting the design variables as a column vector $x = (x_1, x_2, \dots, x_N)^T$, the objective function as a scalar quantity $f(x)$, J inequality constraints as $g_j(x) \geq 0$ and K equality constraints as $h_k(x) = 0$, we write the NLP problem:

$$\begin{array}{ll} \text{Minimize } f(x) & \text{Subject to,} \\ g_j(x) \geq 0 & j = 1, 2, 3, \dots, J; \\ h_k(x) = 0 & k = 1, 2, 3, \dots, K; \\ x_i^{(L)} \leq x_i \leq x_i^{(U)} & i = 1, 2, 3, \dots, N. \end{array}$$

3. Optimization Algorithms

The formulation of engineering design problems differ from problem to problem. They are

- (i) Linear terms for constraints and objective function
- (ii) Non linear terms for constraints and objective function.

The terms are not explicit functions of the design variables. No single optimization algorithm which will work in all optimization problems equals efficiently.

For the sake of clarity, the optimization algorithms are classified into a number of groups, which are now briefly discussed.

(a) Single-variable optimization algorithms.

These algorithms are classified into two categories

- i. Direct methods
- ii. Gradient based methods

Direct methods do not use any derivative information of the objective function; only objective function values are used to guide the search process. However, gradient-based methods use derivative information (first and/ or second order) to guide the search process.

Although engineering optimization problems usually contain more than one variable, single-variable optimization algorithms are mainly used as unidirectional search methods in multivariable optimization algorithms.

(b) Multi- variable optimization algorithms.

These algorithms demonstrate how the search for the optimum point progresses in multiple dimensions. Depending on whether the gradient information is used or not used, these algorithms are also classified into direct and gradient-based techniques.

(c) Constrained optimization algorithms.

These algorithms use the single variable and multivariable optimization algorithms repeatedly and simultaneously maintain the search effort inside the feasible search region. These algorithms are mostly used in engineering optimization problems.

(d) Specialized optimization algorithms.

Two of these algorithms - integer programming and geometric programming - are often used in engineering design problems. Integer programming methods can solve optimization problems with integer design variables. Geometric programming methods solve optimization problems with objective functions and constraints written in a special form.

(e) Non-traditional optimization algorithms.

There are two algorithms which are nontraditional, these are:

- a) Genetic algorithms
- b) Simulated annealing.

4.0 Single-variable optimization algorithms

The algorithms described in this section can be used to solve minimization problems of the following type:

Minimize $f(x)$

Where $f(x)$ is the objective function and x is a real variable. The purpose of an optimization algorithm is to find a solution x , for which the function $f(x)$ is minimum.

4.1 Optimality criteria

There are three different types of optimal points are:

(i) Local Optimal point:

A point or solution x^* is said to be a local optimal point, if no point in the neighbourhood has a function value smaller than $f(x^*)$.

(ii) Global Optimal point:

A point or solution x^{**} is said to be a global optimal point, if no point in the entire search space has a function value smaller than $f(x^{**})$.

(iii) Inflection point:

x^* is an inflection point if

$f(x^*)$ increases locally as x^* increases & decreases locally as x^* reduces

or $f(x^*)$ decreases locally as x^* increases and increases locally as x^* decreases

Let the objective function $f(x)$ is the chosen search space

$f'(x)$ and $f''(x)$ are first and second derivatives

A point x is a minimum if $f'(x) = 0$ & $f''(x) > 0$.

If $f'(x) = 0$, the point is either a minimum, a maximum or an inflection point

Suppose at point x^* , the first derivative is zero and the first non-zero higher order derivative is denoted by n , then

- If n is odd, x^* is an inflection point
- If n is even, x^* is a local optimum
 - (i) If the derivative is +ve, x^* is a local minimum
 - (ii) If the derivative is -ve, x^* is a local maximum

Example: 4

Consider $f(x) = x^3$, optimal point $x = 0$ as shown in Fig.6.

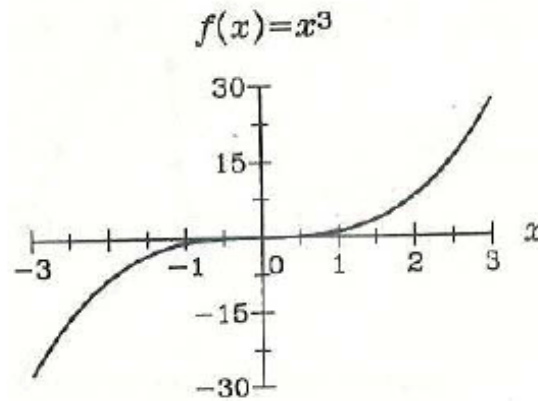


Fig.6. The function $f(x) = x^3$

From the figure, we can see that point $x = 0$ is an inflection point as
 $f(x)$ increases for $x \geq 0$
decreases for $x \leq 0$

Using the sufficient conditions

$$f'(x=0) = 3x^2 \Big|_{x=0} = 0$$

$$f''(x=0) = 6x \Big|_{x=0} = 0$$

$$f'''(x=0) = 6 \Big|_{x=0} = 6(\text{Nonzero value})$$

Third derivative, $n = 3$ is odd, hence $x = 0$ is an inflection point.

Example : 5

Consider $f(x) = x^4$, optimal point $x = 0$ as shown in Fig.6a.

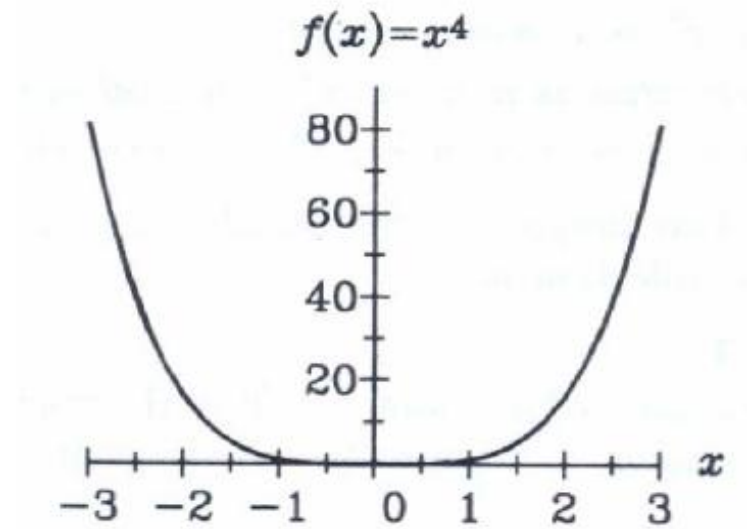


Fig.6a. The function $f(x) = x^4$

From the figure, we can see that the point $x = 0$ is a minimal point.
Using sufficient conditions

$$f'(x=0) = 4x^3 \Big|_{x=0} = 0$$

$$f''(x=0) = 12x^2 \Big|_{x=0} = 0$$

$$f'''(x=0) = 24x \Big|_{x=0} = 0$$

$$f^{(4)}(x=0) = 24 \Big|_{x=0} = 24(\text{Nonzero value})$$

Fourth order derivative is positive, $n = 4$ is even, hence $x = 0$ is a local minimum point.

4.2 Bracketing Methods:

The minimum of a function is found in two phases. Initially an approximate method is used to find a lower and an upper bound of the minimum. Next, a sophisticated technique is used to search within these two limits to find the optimal solution.

(a) *Exhaustive search method*

It is the simplest of all search methods. The optimum of a function is bracketed by calculating the function values at a number of equally spaced points(Fig.7).

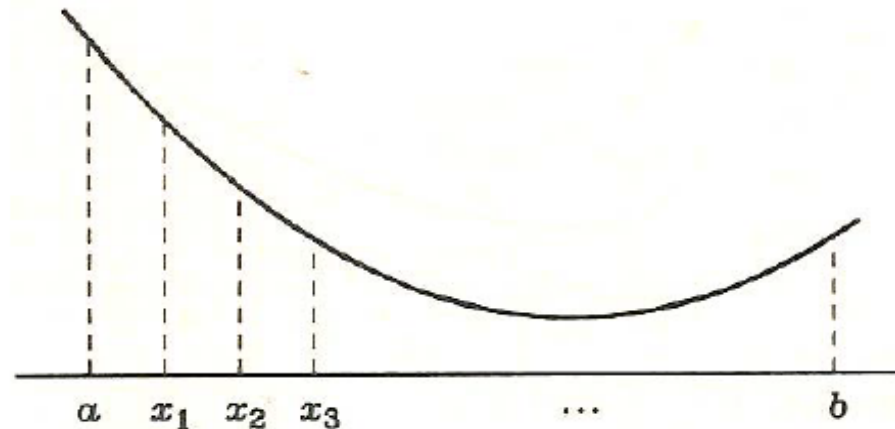


Fig.7 The exhaustive search method that uses equally spaced points

Usually the search begin from a lower bound on the variable and three consecutive function values are compared at a time based on the assumption of unimodality of the function. Based on the outcome of comparison, the search is either terminated or continued by replacing one of the three points with a new point.

Algorithm:

Step 1.

Set

$$x_1 = a, \Delta x = (b - a) / n \text{ (} n \text{ is number of intermediate points)}$$

$$x_2 = x_1 + \Delta x, x_3 = x_2 + \Delta x$$

Step 2

If $f(x_1) \geq f(x_2) \leq f(x_3)$, the minimum point lies between (x_1, x_3) .

Hence, **terminate.**

Else $x_1 = x_2, x_2 = x_3, x_3 = x_2 + \Delta x$

Step 3

Is $x_3 \leq b$? If yes, goto Step 2.

Else no minimum exists in (a,b) or a boundary point (a or b) is the minimum point.

Example:6

Minimize $f(x) = x^2 + 54/x$ in the interval $(0,5)$

A plot of function is shown in Fig.8

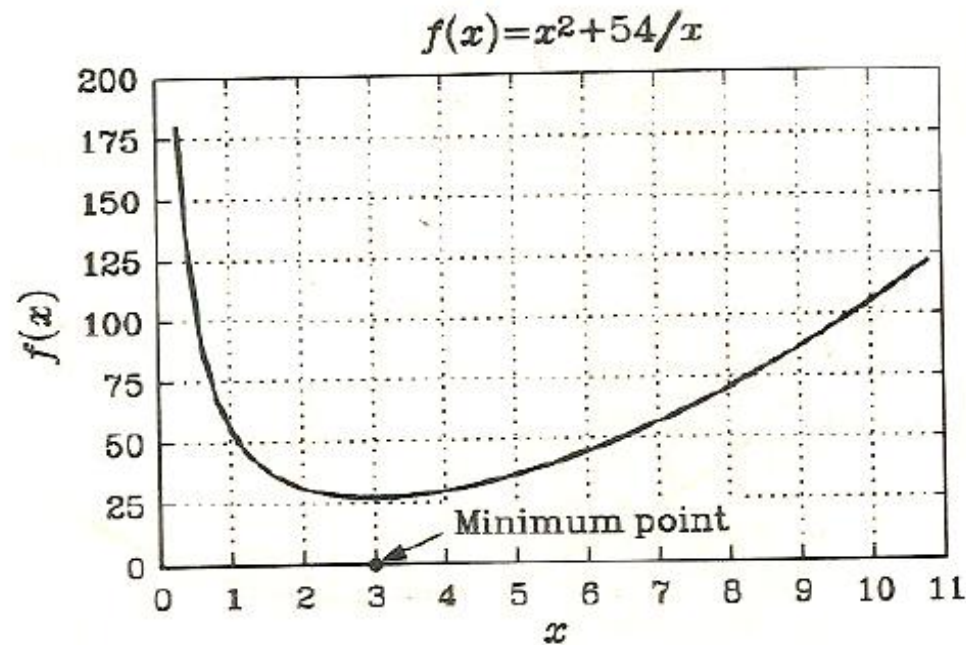


Fig.8 The unimodal, single-variable function used in the exercise problems.

The plot shows that the minimum lies at $x = 3$

$$f(3) = 27$$

$$f'(3) = 0$$

$$f''(3) = 6$$

According to sufficiency conditions, $x = 3$ is a local minimum.

Now consider $n = 10$ for exhaustive search.

Step 1

According to the parameter chosen,

$$x_1 = a = 0 \text{ and } b = 5$$

$$\therefore \Delta x = (5 - 0) / 10 = 0.5$$

We set

$$x_2 = 0 + 0.5 = 0.5$$

and

$$x_3 = 0.5 + 0.5 = 1.0$$

Step 2 Computing function values at various points, we have

$$f(0) = \infty, \quad f(0.5) = 108.25 \quad f(1.0) = 55.00$$

$$f(x_1) > f(x_2) > f(x_3)$$

Thus, the minimum does not lie between (0,1)

$$\therefore \text{Set } x_1 = 0.5, \quad x_2 = 1.0, \quad x_3 = 1.5$$

$$f(1.5) = 38.25$$

$$f(x_1) > f(x_2) > f(x_3) \text{ and minimum does not lie between } (0.5, 1.5)$$

Repeat the process till $f(2.5) = 27.85$

$$f(3.0) = 27.00$$

$$f(3.5) = 27.68$$

$$f(x_1) > f(x_2) < f(x_3)$$

Solution lies between (2.5, 3.5).

The accuracy solution $2(a-b)/n = 2(5-0)/10 = 1.0$

If more accurate solution is required, divide into more number of parts by increasing n .

(b) Bounding phase method

Step 1

Choose an initial guess $x^{(0)}$ and an increment Δ . Set $k = 0$

Step 2

If $f(x^0 - |\Delta x|) \geq f(x^0) \geq f(x^0 + |\Delta x|)$, then Δ is +ve

$f(x^0 - |\Delta x|) \leq f(x^0) \leq f(x^0 + |\Delta x|)$, then Δ is -ve

Else goto Step 1

Step 3

Set $x^{(k+1)} = x^{(k)} + 2^k \Delta$.

Step 4

If $f(x^{(k+1)}) < f(x^{(k)})$, set $k = k+1$ and goto Step 3

Else, the minimum lies in the interval $(x^{(k-1)}, x^{(k+1)})$ and terminate

If Δ is large, accuracy is poor.

Example 7

Find minimum of $f(x) = x^2 + \frac{54}{x}$ using bounding phase method

1. Choose an initial guess $x^{(0)} = 0.6$ and increment $\Delta = 0.5$ set $k = 0$.
2. Calculate three function values

$$f(x^{(0)} - |\Delta|) = f(0.6 - 0.5) = 540.010$$

$$f(x^{(0)}) = f(0.6) = 90.360$$

$$f(x^{(0)} + |\Delta|) = f(0.6 + 0.5) = 50.301$$

We observe that

$$f(0.1) > f(0.6) > f(1.1) \quad \therefore \text{set } \Delta = +0.5$$

3. Next guess: $x^{(1)} = x^{(0)} + 2^0 \Delta = 1.1$
4. $f(x^{(1)}) = 50.301 < f(x^{(0)}) \therefore$ set $k = 1$ and goto Step 3

3. Next guess: $x^{(2)} = x^{(1)} + 2^1 \Delta = 1.1 + 2(0.5) = 2.1$

4. $f(x^{(2)}) = 30.124 < f(x^{(1)})$ $\therefore$ set $k = 2$ and goto Step 3

3. Next guess: $x^{(3)} = x^{(2)} + 2^2 \Delta = 2.1 + 4(0.5) = 4.1$

4. $f(x^{(3)}) = 29.981 < f(x^{(2)})$ $\therefore$ set $k = 3$ and goto Step3.

3. Next guess:

4. $f(x^{(4)}) = 72.277 > f(x^{(3)})$. Thus terminate with interval (2.1, 8.1)

with $\Delta = 0.5$, the obtained bracketing is poor. Functions evaluated are 7.

$\Delta = 0.001$, the obtained interval is (1.623, 4.695). Functions evaluated are 15.

4.3. Region elimination methods:

Once the minimum point is bracketed, a more sophisticated algorithm is used to improve the accuracy of the solution. Region elimination methods are used for this purpose. The fundamental rule for region elimination method is as follows:

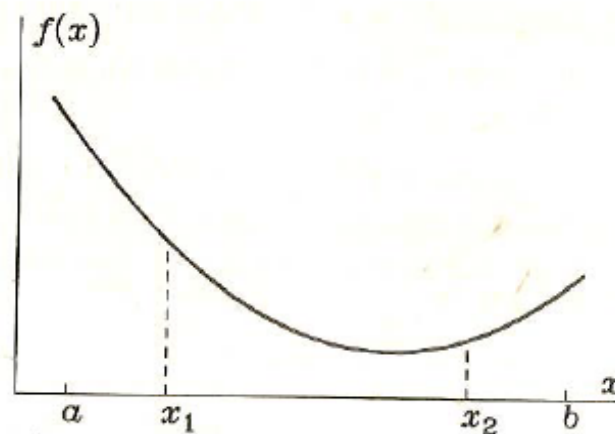


Fig.9. A typical single-variable unimodal function with function values at two distinct points

Consider a unimodal function drawn in Fig.9. The two points x_1 and x_2 lie in the interval (a,b) and satisfy $x_1 < x_2$. For minimization, the following conditions apply

- If $f(x_1) > f(x_2)$ then the minimum does not lie in (a, x_1)
- If $f(x_1) < f(x_2)$ then the minimum does not lie in (x_2, b)
- If $f(x_1) = f(x_2)$ then the minimum does not lie in (a, x_1) and (x_2, b)

(a) Interval halving method:

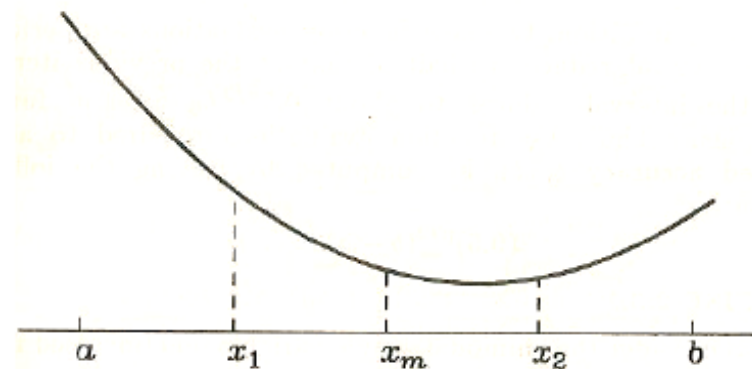


Fig.10 Three points x_1 , x_m , and x_2 used in the interval halving

Fig.10 shows a region in the interval (a, b) . Three points divide the search space into four regions. The fundamental rule for region elimination is used to eliminate a portion of search space based on function values at three chosen points.

x_1 , x_m , x_2 are three search points. Two of the function values are compared at a time and some region is eliminated. The three possibilities are

- (i) If $f(x_1) < f(x_m)$ minimum cannot lie beyond x_m and reduce the interval from (a, b) to (a, x_m) search space is reduced by 50 percent.

(ii) If $f(x_1) > f(x_m)$, minimum cannot lie in the interval (a, x_1) . The point x_1 is $\frac{1}{4}$ in search space, hence reduction is 25 percent.

Then compare x_2 and x_m to eliminate further 25 percent of search space. Continue the process till small enough interval is found.

Algorithm:

1. Choose lower bound a and upper bound b and a small value ϵ for desired accuracy. $x_m = (a+b)/2$, $L_0 = L = b-a$. Compute $f(x_m)$
2. Set $x_1 = a + \frac{L}{4}$, $x_2 = b - \frac{L}{4}$. Compute $f(x_1)$ and $f(x_2)$
3. If $f(x_1) < f(x_m)$, set $b = x_m$, $x_m = x_1$. Goto Step 5, else goto Step 4
4. If $f(x_2) < f(x_m)$, set $a = x_m$, $x_m = x_2$. Goto Step 5, else $a = x_1$, $b = x_2$; goto Step 5
5. Calculate $L = b - a$. If $|L| < \epsilon$, terminate. Else goto step 2.

Fibonacci number

- a series of numbers in which each number (*Fibonacci number*) is the sum of the two preceding numbers. The simplest is the series 1, 1, 2, 3, 5, 8, etc.

(a) Fibonacci search method:

The search interval is reduced according to Fibonacci numbers.

$$F_n = F_{n-1} + F_{n-2} \quad \text{where } n=2,3,4,\dots \text{ and } F_0 = F_1 = 1$$

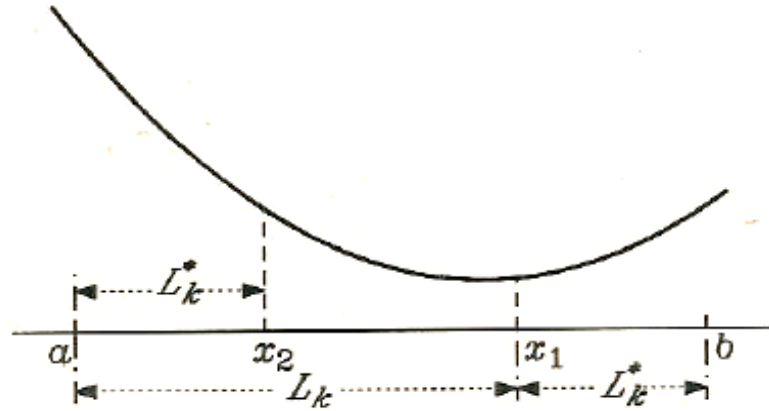


Fig.11. Fibonacci search points(x_1 and x_2)

Algorithm:

1. Choose a lower bound a and an upper bound b . Set $L = b-a$. Assume the desired number of function evaluations to be n . Set $k=2$
2. Compute $L_k^* = (F_{n-k+1}/F_{n+1})L$ Set $x_1 = a + L_k^*$ and $x_2 = b - L_k^*$

3. Compute one of $f(x_1)$ or $f(x_2)$, which was not evaluated earlier. Use the fundamental region elimination rule to eliminate a region. Set new **a** and **b**.
4. Is $k = n$? If no, set $k = k+1$ and goto Step 2. Else terminate.

(c) Golden section search method

Difficulties with Fibonacci search method:

- (i) Fibonacci numbers have to be calculated and stored.
- (ii) At every iteration the proportion of the eliminated region is not the same.

In golden section search method, the search space (a, b) is first linearly mapped to a unit interval search space (0,1). Two points at τ from either end of search space are chosen so that at every iteration the eliminated region is $(1-\tau)$ to that in the previous iteration (Fig.12). This can be achieved by equating

$$(1-\tau) = (\tau \times \tau) \text{ This yields the golden number } \tau = 0.618$$

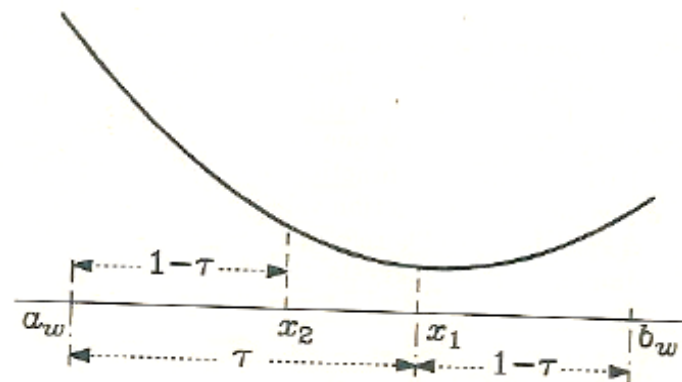


Fig.12. The points (x_1 and x_2) used in the golden section

Algorithm:

1. Choose a lower bound a and an upper bound b . Choose a small value ϵ . Normalize the variable x by using the equation $w = \frac{(x-a)}{(b-a)}$. Thus $a_w = 0, b_w = 1$, and $L_w = 1$. Set $k=1$.
2. Set $w_1 = a_w + (0.618)L_w$ and $w_2 = b_w - (0.618)L_w$. Compute $f(w_1)$ or $f(w_2)$, depending on whichever of the two was not evaluated earlier. Use the fundamental region elimination rule to eliminate a region. Set new a_w and b_w .
3. Is $|L_w| < \epsilon$? If no, set $k=k+1$, go to Step 2. Else terminate.

4.4 Gradient based methods

Despite the difficulty of finding the derivatives, they are popular because of their effectiveness.

(a) Newton- Raphson method

Considering Taylor series expansion $x^{(n+1)} = x^{(n)} - \frac{f'(x^{(n)})}{f''(x^{(n)})}$

Algorithm:

1. Choose initial guess $x^{(1)}$ and small value ϵ . Set $k = 1$. Compute $f'(x^{(1)})$.
2. Compute $f''(x^{(k)})$.
3. Calculate $x^{(k+1)} = x^{(k)} - \left(\frac{f'(x^{(k)})}{f''(x^{(k)})} \right)$ Compute $f'(x^{(k+1)})$.
4. If $|f'(x^{(k+1)})| < \epsilon$, Terminate. Else $k = k+1$ and go to Step 2.

Convergence will depend on initial guess value and nature of the objective function.

Example 8

Minimize $f(x) = x^2 + \frac{54}{x}$ using Newton-Raphson method.

1. Choose initial guess $x^{(1)} = 1$. and $\epsilon = 10^{-3}$, $k = 1$

$$f'(x^{(n)}) = \frac{f(x^{(n)} + \Delta x^{(n)}) - f(x^{(n)} - \Delta x^{(n)})}{2\Delta x^{(n)}}$$

$$f''(x^{(n)}) = \frac{f(x^{(n)} + \Delta x^{(n)}) - 2f(x^{(n)}) + f(x^{(n)} - \Delta x^{(n)})}{(\Delta x^{(n)})^2}$$

$\Delta x^{(n)}$ is parameter, usually taken a small value (0.001).

$$f'(x^{(n)}) = -52.005$$

$$f''(x^{(n)}) = 110.011$$

2.

$$x(2) = x(1) - \frac{f'(x^{(1)})}{f''(x^{(1)})}$$

$$= 1 - \left(\frac{-52.005}{110.011} \right) = 1.473$$

The iterative values of x, for various k's are tabulated below.

k	$x^{(k)}$	$f'(x^{(k)})$	$f''(x^{(k)})$
1	1	-52.005	110.011
2	1.473	-21.944	35.796
3	2.086	-8.239	13.899
4	2.679	-2.167	
-			
7	3.0001	$-4(10)^{-8}$	

f' is small enough to terminate.

Thank You All...